

Document	Detail
Course	GIS Applications in Telecommunications
Day	Day 5 — Visualization, Reporting & Capstone
Version	3.0
Depends on	Days 1–4 outputs (or Day 5 integrated package if prior GDBs unavailable)
Primary tools	Share Web Layer, ArcGIS Online Map Viewer, Dashboards (Indicator/Gauge/Serial/List), ArcGIS StoryMaps
Data folder	Day5_Visualization_Capstone/data/

1. Learning Objectives

By the end of Day 5, participants will be able to:

- Package Days 2–4 telecom GIS results into clean, publishable layers.
- Publish feature layers / web layers to ArcGIS Online and configure pop-ups.
- Build an interactive web map with layer groups for coverage, geo-marketing, fiber, and microwave.
- Create an executive Dashboard using indicators, gauges, serial charts, and lists.
- Author a stakeholder Story Map that narrates the capstone recommendations.
- Deliver a short integrated capstone presentation with clear asks for leadership.

2. Day Agenda

Time	Session	Outcome
09:00–09:30	Capstone framing + data inventory	Team roles & storyline agreed
09:30–10:45	Lab 1 — Integrate/clean layers + static layouts	Publish-ready GDB + 1 overview layout
10:45–11:00	Break	—

11:00–12:15	Lab 2 — Publish to ArcGIS Online + Web Map	Web map with pop-ups/filters
12:15–13:15	Lunch	—
13:15–14:30	Lab 3 — Executive Dashboard	Live dashboard widgets working
14:30–14:45	Break	—
14:45–15:30	Lab 4 — Story Map + recommendations	Narrative capstone ready
15:30–16:00	Team presentations + quiz	Day 5 / course close-out

3. Capstone Storyline (MetroTel)

MetroTel leadership wants one place to see: (1) coverage gaps, (2) high-value and at-risk customers, (3) whether fiber/microwave can support expansion, and (4) prioritized decisions.

- Defend CBD high-ARPU customers (Day 3).
- Expand West / Tech Park / New Town (Day 3 suitability + Day 4 transport).
- Fix peri-urban coverage/QoS issues (Day 2 + churn).
- Choose fiber where microwave LOS is blocked (Day 4).

4. Datasets — Synthetic (lab) + Real (optional)

4.1 Synthetic integrated package

File	Type	Use
ExecZones_Integrated.geojson	Polygons + KPIs	Main dashboard/map geography
ExecZone_KPI_Dashboard.csv	Zone KPI table	Indicators, gauges, charts
Monthly_Executive_KPI_Timeseries.csv	Time series	Serial chart trends
Capstone_CellSites.csv	Sites	Coverage context / candidates
Capstone_KeyNodes.csv	Fiber/MW key nodes	Transport context
Capstone_Decisions.csv	Decision list	List widget + story actions
StoryMap_Outline.csv	Narrative outline	Story Map sections
Dashboard_Widget_Spec.csv	Widget blueprint	Lab 3 build sheet
AGO_Publishing_Checklist.csv	Process checklist	Publishing QA
Popup_HTML_Snippet.txt	Pop-up template	Web map config
Day5_AllTables.xlsx	Excel workbook	Instructor quick view

REAL_DATA_SOURCES.txt	Notes	Replace with real/anonymized ops data
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NOTE: If teams still have Day 2–4 GDBs, publish those authoritative layers instead and use Day 5 tables for dashboard KPIs only.

4.2 Optional REAL / operational data

- Anonymized zone aggregates from CRM/billing (ARPU, churn) — no PII.
- Coverage and site layers from RF planning (Day 2).
- As-built fiber/MW from transmission GIS (Day 4).
- Org basemap + Living Atlas demographics if licensed.

5. Project Setup

- Create ArcGIS Pro project MetroTel_Day5_Capstone.aprx and GDB MetroTel_Day5.gdb
- Folder connection to Day5_Visualization_Capstone/data
- Import ExecZones_Integrated.geojson → ExecZones
- XY Table To Point: Capstone_CellSites.csv → CellSites; Capstone_KeyNodes.csv → KeyNodes
- Add tables: ExecZone_KPI_Dashboard, Monthly_Executive_KPI_Timeseries, Capstone_Decisions
- Join ExecZones to KPI table on DIST_ID if fields not already present; export ExecZones_KPI
- Symbolize PRIORITY with distinct colors (P1 red/orange, P2 blue, P3 amber, P4 gray)
- Confirm ArcGIS Online / Enterprise org login works (Share ribbon available)

CHECKPOINT: Map shows 12 zones by priority, sites, and key transport nodes.

6. Lab 1 — Integrate Results & Static Visualization

Objective: Produce a clean, presentation-grade static overview before web publishing.

- Create layer group structure in Pro:
- 01_Coverage_Sites (CellSites)
- 02_GeoMarketing_Zones (ExecZones_KPI)
- 03_Transport_Nodes (KeyNodes)
- 04_Decisions (optional table join/labels)
- Configure pop-ups in Pro using fields DIST_NAME, PRIORITY, AVG_ARPU, AVG_CHURN, COV_4G_PCT, FIBER_STATUS, MW_BACKHAUL
- Insert Layout 16:9 → title “MetroTel Capstone Overview”
- Add legend, north arrow, scale, date, data note: Synthetic training data / Version 3
- Export PDF: Day5_Capstone_Overview.pdf

- Optional: export zone summary table to Excel for annex

CHECKPOINT: One static overview PDF exported; layer groups organized.

7. Lab 2 — Publish Web Layers & Build Web Map

Objective: Move from desktop GIS to ArcGIS Online dynamic maps.

- In Pro: Share → Web Layer → Publish ExecZones_KPI as feature layer MetroTel_ExecZones_v3
- Publish CellSites as MetroTel_CellSites_v3
- Publish KeyNodes as MetroTel_KeyNodes_v3
- Publish Capstone_Decisions as table or hosted table MetroTel_Decisions_v3
- Optionally publish Monthly KPI table for dashboard charts
- In ArcGIS Online Map Viewer: create web map MetroTel_Capstone_WebMap
- Add the hosted layers; set basemap to Dark Gray Canvas or Topographic
- Configure pop-ups with Arcade or the HTML snippet from Popup_HTML_Snippet.txt
- Add filters: PRIORITY, FIBER_STATUS, STATUS (sites)
- Save bookmarks: CBD Defend, West Expand, Peri-Urban Fix
- Set sharing to Organization or Training Group

TIP: If publishing credits/time are limited, one combined feature service with multiple layers is acceptable.

NOTE: If AGOL is unavailable, use Pro as a fallback demo and still design dashboard/story wireframes from the specs.

CHECKPOINT: Web map opens in AGOL with working pop-ups and at least two filters/bookmarks.

8. Lab 3 — Executive Dashboard

Objective: Build an interactive dashboard with indicators, gauges, charts, and lists.

- ArcGIS Online → Content → Create → Dashboard → MetroTel_Executive_Dashboard_v3
- Add Map element pointing to MetroTel_Capstone_WebMap; enable legend + layer selectors
- Add Indicator: Total Subscribers = Sum(SUBSCRIBERS) from zones layer/table
- Add Indicator: Mean AVG_ARPU
- Add Gauge: Mean COV_4G_PCT (target/reference 85)
- Add Gauge: Mean AVG_CHURN (target/reference max 0.35; lower is better)
- Add Serial Chart: Monthly_Executive_KPI_Timeseries (MONTH vs SUBSCRIBERS and ARPU_USD)
- Add Pie/Category chart: count of zones by PRIORITY

- Add List: Capstone_Decisions filtered where STATUS is not Approved; show TITLE, OWNER, PRIORITY
- Enable map selection filtering so clicking a zone updates indicators (if configured)
- Apply consistent theme/colors; title "MetroTel Network & Market Command View"
- Save & share with same group as the web map

CHECKPOINT: Dashboard shows ≥ 2 indicators, ≥ 1 gauge, ≥ 1 serial chart, ≥ 1 list, and the map.

9. Lab 4 — Story Map & Capstone Ask

Objective: Package the narrative for executives using ArcGIS StoryMaps.

- Create → StoryMap titled MetroTel Capital Region — GIS Capstone
- Follow StoryMap_Outline.csv sections:
 - Cover + brief problem statement
 - Map: coverage/sites
 - Map: revenue / ARPU
 - Map: churn / at-risk
 - Map: fiber & microwave readiness
 - Map: PRIORITY decisions
 - Closing panel: leadership asks
- Embed the web map and/or dashboard where useful (sidecar / map tour / embed)
- Add 3 explicit asks:
 - Approve west fiber build to FDH-04 (prefer over blocked MW).
 - Authorize candidate site evaluation in peri-urban/new-town gaps.
 - Fund CBD retention program for high-ARPU defend zones.
- Add credits/methodology note: Days 2–4 methods summarized in 5 bullets
- Share Story Map to org/group; test on phone/projector width

CHECKPOINT: Story Map has cover + ≥ 4 map sections + closing asks; link works.

10. Capstone Presentation Guide (10–15 min)

Minute	What to show	Message
0–2	Story cover / problem	Why one integrated GIS view matters

2–5	Dashboard KPIs	Subs, ARPU, coverage, churn at a glance
5–8	Web map bookmarks	Defend / Expand / Fix zones
8–11	Transport readiness	Fiber vs MW decision for west/tech park
11–14	Decision list	Owners, priorities, next actions
14–15	Asks	Approvals required from leadership

11. Expected Outputs Checklist

#	Deliverable	Done?
1	MetroTel_Day5_Capstone.aprx + GDB	
2	Static overview layout PDF	
3	Hosted layers in ArcGIS Online (zones/sites/nodes/decisions)	
4	Web Map with pop-ups, filters, bookmarks	
5	Dashboard with indicator/gauge/serial/list/map	
6	Story Map narrative with leadership asks	
7	Capstone presentation (live or slides + links)	
8	Sharing set to training org/group + metadata note	

12. Instructor Answer Key — Sample Capstone Messages

- P1-Expand typically includes West Expansion / Tech Park / New Town in the packaged seed.
- CBD is a defend / high-ARPU story with lit fiber.
- Peri-Urban North is a fix-coverage / QoS story with weak fiber presence.
- Transport recommendation: do not rely on planned MW into west without LOS clearance; advance planned fiber.
- Dashboard should make P1 count and churn/coverage gauges visible without opening Pro.

13. Troubleshooting

Problem	Likely cause	Fix
Share Web Layer disabled	Not logged into org / no publish privilege	Login; request publisher role
Dashboard chart empty	Wrong layer/table or field types	Use hosted table; set MONTH as category

Pop-ups show field names only	No configure pop-up	Use custom HTML/Arcade snippet
Map/dashboard not visible to teammates	Sharing level private	Share layers+map+dashboard+story to same group
Too much detail / slow map	Publishing full raw subscribers	Use zone aggregates for AGOL; keep points local
No AGOL in classroom	Network/policy limits	Build offline wireframes + Pro layouts; publish later

14. Quick Quiz

- Name three ArcGIS Online products used on Day 5.
- Which widget best shows a single KPI vs target (e.g., coverage %)?
- Why publish zone aggregates instead of raw customer points for executives?
- What should a Story Map include that a Dashboard usually does not?
- Give one example of a leadership “ask” from this capstone.

Suggested answers:

- 1) Web Map / hosted feature layers, Dashboards, StoryMaps.
- 2) Gauge (or Indicator with reference).
- 3) Privacy, performance, clearer decisions at market level.
- 4) Narrative sequence and persuasive recommendations/asks.
- 5) e.g., Approve west fiber to FDH-04; fund CBD retention; authorize candidate sites.

15. Program Close-Out

Days 1–5 chain: fundamentals & GDB → coverage/site selection → geo-marketing → fiber/microwave → executive visualization. Encourage teams to document data refresh cadence (weekly KPI table update) and keep a single AGOL group as the operational home for MetroTel GIS products.

16. Version History

Version	Changes
v1	Draft Day 5 visualization outline
v2	Added dashboard/story sequence
v3	Full manual with integrated MetroTel capstone datasets, AGOL publish steps, dashboard widget build, Story Map outline, presentation guide, checkpoints, quiz, real-data/privacy notes